FUNCTION

1. Let A and B be sets with cardinalities m and n respectively. Find the number of

A). Functions B). One-one functions C). Many one function

D).Onto functions E). Into functions

2. Suppose X and Y are sets and |X| and |Y| are their respective cardinalities. It is given that there are exactly 97 functions from X to Y. from this one can conclude that

A). |X| = 1, |Y| = 97 B). | X |= 97, |Y| = 1

C). |X| = 97, |Y| = 97 D). None of the above

3. Let R denotes the set of real numbers. Let $f: R \times R \to R \times R$ be a bijective function defined by $f(x,y) = (x+y, x-y)$. The inverse function of f is given by

A). $f^{-1}(x,y) = \left(\frac{1}{x+y}, \frac{1}{x-y}\right)$ B). $f^{-1}(x,y) = (x-y, x+y)$

C). $f^{-1}(x,y) = \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$ D). $f^{-1}(x,y) = \left(2(x-y), 2(x+y)\right)$

4. Let F be the set of one-to-one functions from set $\{1,2,3,.....n\}$ to the set $\{1,2,3,.....m\}$. Where $m \geq n \geq 1$.

A). How many functions are members of F?

B). How many functions f in F satisfy the property $f(i) = 1$ for some $i, 1 \leq i \leq n$?

C). How many functions f in F satisfy the property $f(i) < f(j)$ for all $1 \leq i \leq j \leq n$?

5. Let $f: A \to B;$ be a function, and let E and F be subsets of A. Consider the following statements about images. $S1: f(E \cup F) = f(E) \cup f(F)$ $S2: f(E \cap F) = f(E) \cap f(F)$

Which of the following is true about S1 and S2?

A). Only S1 is correct B). Only S2 is correct

C). Both S1 and S2 are correct D). None of S1 and S2 is correct

6. Consider the function $h: N \times N \to N$ so that $h(a,b) = (2a+1)2^b - 1$, where $N = \{0,1,2,3.....\}$ is the set of natural numbers.

A). Prove that the function h is an injection (one-one).

B). Prove that it is also a Surjection (onto)

7. Let $f: A \to B$ be an injective (one-to-one) function. Define $g: 2^A \to 2^B$ as:

$g(C) = \{f(x) \mid x \in C\}$, For all subsets C of A

Define $h: 2^B \to 2^A$ as: $h(D) = \{x \mid x \in A, f(x) \in D\}$, for all subsets D of B.

Which of the following statements is always true?

A). $g(h(D)) \subseteq D$ B). $g(h(D)) \supseteq D$

C). $g(h(D)) \cap D = \varnothing$ D). $g(h(D)) \cap (B - D) \neq \varnothing$

8. Let $\Sigma = \{a, b, c, d, e\}$ be an alphabet. We define an encoding scheme as follows:

$g(a) = 3, g(b) = 5, g(c) = 7, g(d) = 9, g(e) = 11$. Let p_i denote the ith prime number $(p_1 = 2)$. For a non-empty string $s = a_1 a_2 \ldots\ldots a_n$, where each $a_i \in \Sigma$, define $f(s) = \prod_{i=1}^{n} p_i^{g(a_i)}$. For a non-empty sequence $< s_1, s_2 \ldots\ldots s_n >$ of strings from Σ^+, define $h(< s_1 s_2 \ldots s_n >) = \prod_{i=1}^{n} p_i^{f(s_i)}$ .IF

A). $s_1 = a, s_2 = b, s_3 = c, s_4 = d, s_5 = e;$ B). $s_1 = a, s_2 = a, s_3 = a, s_4 = a;$

C). $s_1 = ab, s_2 = bc, s_3 = abc;$ D). $s_1 = c, s_2 = ab;$

Find $h(< s_1 s_2 \ldots s_n >)$ in each case.

9. Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subsets of W. The number of functions from Z to E is:

A). $Z^{2^{xy}}$ B). $Z \times 2^{xy}$ C). $Z^{2^{x+y}}$ D). 2^{xyz}

10. Let $S = \{1, 2, 3, \ldots m\}, m > 3$. Let $X_1, X_2, \ldots\ldots\ldots, X_n$ be subsets of S each of size 3. Define a function f from S to the set of natural numbers as, $f(i)$ is the number of sets X_j that contain the element i. That is $f(i) = |\{j \mid i \in X_j\}$, Then $\sum_{i=1}^{m} f(i)$ is

A). $3m$ B). $3n$ C). $2m+1$ D). $2n+1$

11. How many onto (or Surjective) functions are there from an n-element $(n \geq 2)$ set to a 2-element set?

A). 2^n B). $2^n - 1$ C). $2^n - 2$ D). $2(2^n - 2)$

12. Let S denote the set of all functions $f:\{0,1\}^4 \to \{0,1\}$. Denote by N the number of functions from S to the set $\{0,1\}$. The value of $\log_2 log_2 N$ is _____.

13. Let X and Y be finite sets and $f: X \to Y$ be a function. Which one of the following statements is TRUE?

A).For any subsets A and B of X, $|f(A \cup B)| = |f(A)| + |f(B)|$

B).For any subsets A and B of X, $f(A \cap B) = f(A) \cap f(B)$

C).For any subsets A and B of X, $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$

D).For any subsets S and T of Y, $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

14. Consider the set of all functions $f:\{0,1,2,.....,2014\} \to \{0,1,2,.....,2014\}$ such that $f(f(i)) = i$, for all $0 \le i \le 2014$. Consider the following statements:

P. For each such function it must be the case that for every i, $f(i) = i$.

Q. For each such function it must be the case that for some i, $f(i) = i$.

R. Each such function must be onto.

Which one of the following is CORRECT?

A). P, Q and R are true

B). Only Q and R are true

C). Only P and Q are true

D). Only R is true

15. Total number of onto functions form a set A of 5 elements to a Set B of 4 elements is _____.